[INCH-POUND]
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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

HAT, SUN, REVERSIBLE

The General Services Administration has authorized the use of this commercial item description for all Federal agencies.

1. SCOPE. This commercial item description covers the requirements for a two-color reversible, cotton warp and nylon filling treated with water repellent, wide-brimmed, adjustable size, sun hat.

2. CLASSIFICATION. The hat shall be of one type and size.

3. SALIENT CHARACTERISTICS

3.1 Description. The hat is a lightweight, soft crowned, reversible, two-color design with a stitched brim and a size adjustment strap interlaced through six buttonhole slits at base of the crown (see Figure 1). The brim shall be stitched around the outer edge of brim 1/16 to 1/8 inch from the edge with all plies smooth. The entire brim shall be stitched spirally with the rows of stitching 1/8 to 1/4 inch apart.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may of use in improving this document should be addressed to: Defense Supply Center Philadelphia, Clothing and Textiles Directorate, ATTN: DSCP-CRFD, 700 Robbins Ave, Bldg 6D, Philadelphia, PA 19111-5092.

AMSC N/A FSC 8415

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3.2. Material. The basic material shall be an oxford cloth made of cotton warp and nylon filling treated with a water repellent. The warp yarn shall be 2-ply and the filling yarn shall be single ply, semi-dull filament. The color of the cloth for one side of the crown and brim shall be a Woodland Camouflage Pattern (Light Green 354, Dark Green 355, Brown 356, and Black 357). The color of the cloth for the other side of the crown, brim, and adjustment strap lining shall be Orange 338. All colors of the finished cloth shall match the standard sample when viewed under filtered tungsten lamps that approximate artificial daylight and that have a correlated temperature of 7500 (+/-200 K), with illumination of 100 (+/-20) foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2300 (+/- 200 K). The basic material shall conform to the requirements specified in Table I and Table II.

TABLE I. Basic material physical requirements

Characteristic	Requirement	Test method
Weave	Oxford 1/	Visual
Weight, oz./sq. yd.	4.8 – 5.8	ASTM-D-3776,Opt.C
Yarns per inch (min.)		
Warp	160	ASTM-D-3775
Filling	74	ASTM-D-3775
Breaking strength (lb. Min.)		
Warp	150	ASTM-D-5034
Filling	125	ASTM-D-5034
Water repellency hydrostatic, height, cm., (min.)	35	ASTM-D-751, Procedure 1 with 3 drops
Dynamic absorption (percent, max.)		
Initial	25	AATCC 70
After 15 laundering	25	FTM 5556 and AATCC 70
Spray rating	90, 90, 80	AATCC 22
Air permeability, ($ft.^3/min./ft.^2$), max	4.0	ASTM-D-737
Resistance to organic liquid (n-tetradecane)	No wetting	AATCC 118
Colorfastness to:		
Laundering	3 – 4 2/	AATCC 61, Test 3A
Perspiration	3 – 4	AATCC 15
Crocking	3 – 4	AATCC 8
Light	3 – 4	AATCC 16, Option A
Spectral reflectance, (nanometers)	600 – 860	see Table II
Non-fibrous material, (percentage, max.)	2.0	ASTM-D-629
PH	5.0 – 8.5	ASTM-D-2165

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TABLE I. Basic material physical requirements – continued

Characteristic	Requirement	Test method
Dimensional stability (percentage, max.)		
Warp	2.0	AATCC 96
Filling	2.0	AATCC 96
Seam efficiency, (percentage, max.)	70	ASTM-D-1683

1/ 2 warp ends weaving as one
2/ Equal to or better than the specified rating on AATCC Gray Scale For Color Change and/or Staining when compared to the original unexposed specimen.

TABLE II. Spectral reflectance

	Values (percent)					
	Light Green 354		Dark Green 355 Brown 356		Black 357	
Wavelength						
nanometers	min	max.	min.	max	min	max.
600	8	18	3	10	---	10
620	8	18	3	10	---	10
640	8	18	3	9	---	10
660	8	18	3	12	---	10
680	10	22	3	14	---	10
700	18	40	5	18	---	10
720	22	45	7	20	---	10
740	30	55	12	28	---	10
760	35	65	18	36	---	10
780	40	75	26	44	---	10
800	45	80	34	52	---	10
820	50	86	42	60	---	10
840	55	88	50	68	---	10
860	60	90	56	74	---	10

3.2.1 Brim interlining. The material for the brim interlining shall be either a cotton, plain weave, or a nonwoven consisting of no less than 80% polyester with the remainder cellulose. The interlining shall have a weight of 4.5–7.5 oz./sq. yd when tested in accordance with ASTM-D-3776, Option C, and a non-chlorine retentive resin finish. The color of the cloth shall be either natural or bleached white. The breaking strength shall be a minimum of 55 (warp) and 45 (filling) when tested in accordance with ASTM-D-5034, Option G-T, or Option G-E. The shrinkage of the interlining shall be consistent with the end item.

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3.2.2 Adjustment strap. The material for the adjustment strap shall be a 0.21 oz. per linear yard (minimum), 5/8 inch wide cotton tape, and the color shall be Olive Green 106 or Camouflage Green 483. The lining side shall be cotton warp and nylon filling and the color shall be Orange 338.

3.2.3 Resin impregnating agent. The resin-impregnating agent for treatment of the strap ends shall be either cellulose acetate, cellulose butyrate, or cellulose acetate butyrate.

3.2.4 Snap fasteners. The snap fasteners shall be brass with a black chemical finish, 16-line, male and female complete, consisting of four separate components, pronged ring, socket, stud, and eyelet. Six snap fasteners shall be attached to the adjustment strap. The first socket (female) and first stud (male) shall be positioned 1/2 (+/- 1/4) inch from the end of strap, measured from the center of the fasteners to end of strap. The remaining five sockets shall be positioned 1 (+/- 1/16) inch apart (center to center), and the remaining five studs shall be positioned to correspond with the sockets. The studs shall finish on the tape side (green) and the sockets shall finish on the lining side (orange).

3.2.5 Threads. The sewing threads shall be cotton-covered polyester, or equal Tex 35. The threads shall have a breaking strength of 2.6 lb. (min.) and maximum elongation of 26 percent when tested in accordance with ASTM-D-2256. The color of the threads shall be Olive Drab S-1, C.A. 66022 (for one side of the crown and brim) and Orange TH number 66074 (for the other side of the crown and brim and the adjustment strap lining). Colorfastness shall be consistent with the end item. The finish of the threads shall be non-staining, and non-flame propagating.

3.2.6 Identification label. Each hat shall have an identification label that shall be applied by either marking, direct printing, stamping, or stenciling. The label shall contain an item description, contract number, National Stock Number (NSN), and contractor's name. The label shall be applied to the orange side of the brim, with the bottom of the label positioned 2 (+/- 1/4) inches from the brim edge, measured at the center of the label. The character size of the inscriptions shall be minimum 18 point (Approximately 1/4 inch).

3.2.7 Label/tag. Each item shall be individually bar-coded with a paper tag for personal clothing items. The paper tag shall be standard bleach sulfate having a basic weight of 100 pounds. The paper used for the tags shall have a smooth finish to accept thermal transfer and direct printing. The tags shall have a hole and shall be attached to each item by a fastener, clearly legible and readable by scanner. The bar coding element shall be a 13 digit national stock number (NSN). The bar code type shall be a medium to high code density and shall be located so that it is completely visible on the item when it is packaged and so that it causes no damage to the item.

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3.3. MANUFACTURING REQUIREMENTS

3.3.1 Stitches and seams. All raw edges shall be turned under. All stitching shall be 8 to 12 stitches per inch except buttonholes, which shall have 42 to 48 stitches including tack. All stitch types shall be in accordance with ASTM standards. The brim shall be attached to the crown using stitch type 301. Crown assembly and topstitching shall be stitch type 301 or 401. Adjustment strap and brim shall be stitch type 301.

3.3.2 Buttonholes. The hat shall require six buttonholes. The buttonholes shall be 3/4 to 7/8 inch vertical buttonholes centered in each crown panel (see Figure 1). The bottom of the buttonhole shall be 1/4 to 3/8 inch from the bottom edge of the crown. Buttonholes shall use a 304 zig zag stitch type.

3.4 Patterns. The Government will furnish a set of standard patterns to be used to cut working patterns. The working patterns shall be identical to the Government patterns. Neither the Government patterns nor the working patterns shall be altered in any way, except that additional notches for use during construction are allowed on the working patterns. Also minor modifications are permitted where necessary when using automatic equipment. These modifications shall not alter the dimensional, serviceability, or appearance requirements. The standard patterns provide a seam allowance of 1/8 inch for outer brim edge, 3/8 inch for bottom crown edge, and 1/4 inch for all other seams. Commercial patterns may be used in lieu of the government furnished pattern provided they conform to the cited design and finished measurements.

3.4.1 Pattern parts. The pattern parts and applicable material for cutting the hat parts shall be as specified in Table III

TABLE III. List of pattern parts

Material	Nomenclature	Cut parts
Oxford cloth Woodland Camouflage	 Crown 1/ Brim 2/ Brim alternate 3/	 6 1 2
Oxford cloth Orange	 Crown 1/ Brim 2/ Brim alternate 3/	 6 1 2
Buckram or twill	Brim interlining Brim interlining alternate 4/	1 2

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1/ Use same crown pattern
2/ Use same brim pattern
3/ Use same crown pattern
4/ Use same brim pattern

3.5 Finished measurements. The finished dimensions for the hat shall conform to Table IV. all measurements and tolerances are expressed in inches.

TABLE IV. Finished measurements

Circumference 1/	Crown height 2/	Brim width 3/	Length of strap 4/
24-3/4 +/- 1/4	7-1/4 +/- 1/8	2-3/4 +/- 1/4	25-1/4 +1/4, -0

1/ Remove adjustment strap, measure inside circumference along brim-to-crown joining seam.
2/ Measure through center of one of the side panels from the brim-to-crown joining seam to the crown center seam.
3/ Measure (any place on brim) from brim-to-crown joining seam to the outer edge of brim.
4/ With nominal tension on strap measure from end to end.

4. REGULATORY REQUIREMENTS

4.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally referable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, and shall conform to the producer's own standards and quality assurance practices. The Government reserves the right to require proof of such conformance.

5.2 Visual examination. Each hat shall be examined for the defects listed below.

5.2.1 Defects. Any hole, cut, needle chew, or tear; color not as specified; any part shaded; any spot or stain; raw edges, open seams, thread ends not removed, or loose tension resulting in loose seams; any material defects, distorted parts, seams twisted, puckered or pleated; any component part omitted; any component not as specified; any operation omitted, broken stitches, buttonhole cut length not as specified, ragged edges, label missing, incorrect, or illegible; buttonhole not vertically positioned; measurement of item not as specified; orange thread on camouflage side, green thread on orange side; snap fastener (s) omitted, snap fastener (s) not securely clinched or defective affecting function; snap fastener (s) clinched to tightly, cutting fabric, rough edges on snap fastener (s); bar code label omitted or not readable by scanner; human-readable interpretation (HRI) omitted or illegible; bar code label not

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visible on folded, packaged item; bar code causes damage to the item; any items not packaged in accordance with the contract or purchase order.

5.3 Acceptance criteria. Acceptance criteria shall be as specified in the contract or purchase order.

6. PACKAGING

6.1 Preservation, packing, and marking. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES

7.1 Source of Government documents. Copies of military and Federal documents are available from:

Standardization Documents Order Desk
Bldg. 4D
700 Robbins Avenue
Philadelphia, PA 19111-5096

7.2 Source of non-Government documents

ASTM Test Methods

ASTM Standards Related to Stitches and Seams

(Applications for copies should be addressed to American Society For Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428).

AATCC Test Methods

(Applications for copies should be addressed to American Association of Textile Chemists and Colorists (AATCC), P. O. Box 12215, Research Triangle Park, NC 27709-2215.)

COLOR ASSOCIATION OF THE UNITED STATES

STANDARD COLOR CARD OF AMERICA

(Standard shades may be available from the Color Association of the United States, Inc., 589 8th Avenue, New York, NY 10018. If standard shades are not available from the Color Association, individual color samples may be obtained from the contracting activity or as directed by the contracting activity.)

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MILITARY INTERESTS:

Custodians:
Army – GL
Air Force – 82

Review Activities:
Army – MD
Air Force – 45, 99

CIVIL AGENCY OORDINATING ACTIVITY:
GSA – FSS

PREPARING ACTIVITY:
DLA – CT

Project 8415-0164

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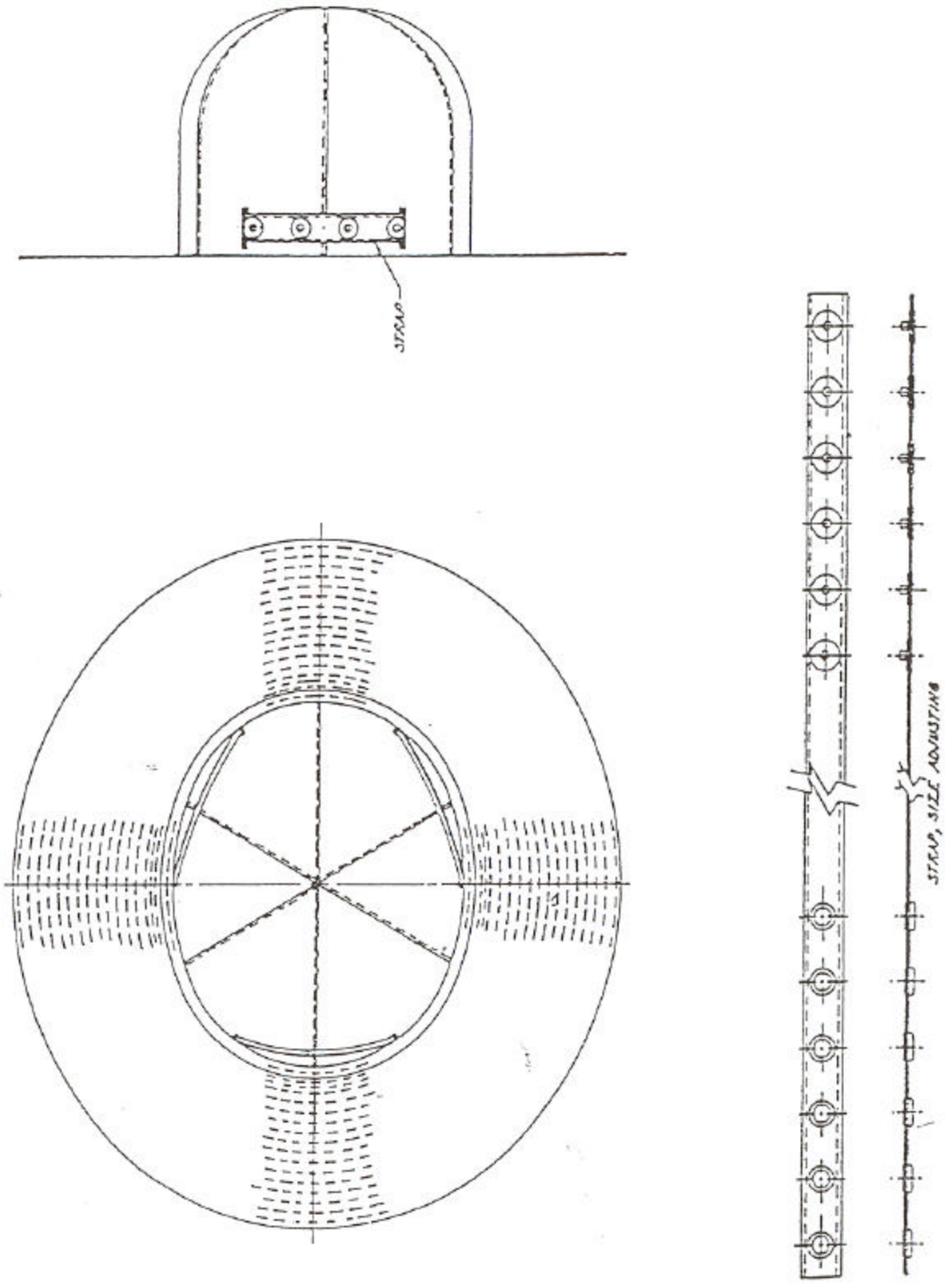


FIGURE 1. Hat, Sun, Reversible

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